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Review Article

Modelling ac voltammetry with MECSim: facilitating simulation–experiment comparisons

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Here we introduce the Monash Electrochemistry Simulator (MECSim) software package that allows most aspects of ac voltammetry to be simulated when a single sine wave or a combination of sine waves are superimposed onto a dc ramp. The MECSim software and companion tool kits can be downloaded free of charge from <http://www.garethkennedy.net/MECSim.html> where there is also instructional documentation. Features accommodated in the simulation model include non-linear background capacitance, uncompensated resistance, Butler–Volmer and Marcus–Hush electron transfer, chemical reactions involving surface confined and solution phase species and stationary or rotating disc electrode mass transport with linear diffusion. Companion tool kits are also provided that allow Fourier transformed ac voltammograms with harmonic analysis to be generated along with other forms of data presentation. The MECSim format in combination with parameter optimisation tool kit also facilitates the implementation of sophisticated forms of experiment–theory comparisons needed to estimate the electrode kinetic and other parameters. MECSim software is computationally efficient and based on flexible code that can be used in cluster computing as well as on a single machine. Typical simulations take seconds to run on a laptop and parameter searches needed to tackle the inverse problem can be achieved efficiently. Capabilities of MECSim are illustrated in this article by comparing simulated results with those obtained experimentally.

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Introduction

In the Monash University Electrochemistry Group, considerable attention has been paid to exploiting advantages of ac voltammetry in mechanistic studies of electrode processes. In this technique, a periodic signal derived from a single sine wave or a combination of sine waves of any amplitude or frequency is superimposed onto the same potential ramp as used in dc voltammetry [1,2^{••},3[•],4^{••},5^{••}].

Dynamic electrochemical techniques that involve current flow are widely used in many branches of science to explore reactions involving electron transfer. The two most widely used versions are probably cyclic voltammetry in its dc form which generates potential–current–time curves and impedance spectroscopy in which it is common for a small amplitude periodic signal containing many frequencies to be superimposed onto a dc potential and data are analysed in the frequency rather than time domain. In ac voltammetry, the technique of interest in this paper, a periodic waveform is superimposed onto the dc ramped potential and the resultant potential–current–time–frequency data can be analysed in ways that exploit the advantages of both dc cyclic voltammetry and impedance spectroscopy.

There are an infinite number of reaction pathways that can take place in a dynamic electrochemical experiment. In order to deduce the mechanism in dc cyclic voltammetry and quantify relevant parameters, it is now common to undertake simulations of the response using in-house software or commercially available packages such as DigiSim [6^{••},7], DigiElch [8[•]], KISSA [9[•],10] and compare these data with experimental results. In impedance spectroscopy, analysis is generally undertaken using equivalent circuits to model the data [11[•],12]. Since the large amplitude version of ac voltammetry has the attributes of both dc cyclic voltammetry and impedance spectroscopy, it is highly sensitive to electrode kinetics, chemical reactions coupled to electron transfer, uncompensated resistance and capacitance, particularly when Fourier transform (FT) methods are used to resolve into the aperiodic (dc) component and a series of ac harmonics which can then be examined independently from the full body of data.

From a theoretical perspective, ac voltammetry is closely related to the dc form, so it should be possible to generate an ac simulation package to encompass a vast array of mechanisms. From a mathematical perspective,

impedance spectroscopy and ac voltammetry in the FT version are identical when the potential is scanned in both forms of experiment and no restrictions are placed on the amplitude (small amplitudes are often used in impedance case to simplify the theory). Thus, a straightforward simulation of an ac voltammogram in terms of using well-established thermodynamic and electrode kinetic parameters could be expected in many situations to replace the less intuitively obvious equivalent circuit forms of analysis based on resistors, capacitors and other electronic elements used in impedance spectroscopy. This voltammetric simulation capacity and other useful features have been implemented in the Monash Electrochemistry Simulator MECSim. However, if the MECSim simulation package is to be used with maximum impact, then mathematical tools such as Fourier transforms used to resolve aperiodic and harmonic components and parameter optimisation tools for assessing the level of agreement between theory and experiment are needed. A collection of such tools is available at <http://www.garethkennedy.net/MECSim.html>. Ideally, use of this software package will be integrated with instrumentation to provide an advanced strategy for bundling experiment and theory and their comparison, as has been achieved at Monash University.

The release of the details of MECSim and accompanying tool kit is aimed at providing users of ac voltammetry the same flexibility in undertaking simulations that is available in dc voltammetry and impedance spectroscopy. The generalised approach adopted allows MECSim to accurately and efficiently model: any combination of ac sinusoids; mass transport at planar, spherical, cylindrical, and rotating disc electrodes; non-linear background capacitance; uncompensated resistance, and any number of interacting homogeneous solution phase chemical reactions coupled to electron transfer. Some reaction schemes involving interaction between species in solution and those fixed onto the electrode surface also can be simulated. The freely available MECSim package consists of an executable file and a text file input for entry of the simulation parameters. Using the text files as input makes it versatile so it is easy to use with scripting languages and in conjunction with cluster and cloud computing, a feature not available with most commercial electrochemical software. Importantly, implementation of powerful automated multi-parameter data optimisation procedures is available [3,13,14*,15*,16,17*,18,19*].

The model that MECSim mimics consists of a user defined, time-dependent potential applied to a working electrode placed in an electrochemical cell with the reference and auxiliary electrodes along with the electroactive chemical species, solvent and electrolyte of interest. In this dynamic electrochemical experiment, contributions from faradaic current associated with electron transfer, capacitive background current and constant uncompen-

sated resistance are assumed to be present. MECSim allows modelling the charge transfer kinetics using either the Butler–Volmer parameterisation based on macroscopic concepts or the more physically realistic Marcus–Hush theory. A non-linear response to the applied potential of the capacitor that generates the background can also be modelled. Such potential-dependent behaviour can be neglected when the faradaic to background current ratio is large [14*,20–22], but otherwise needs to be properly simulated as it contributes to the IR_u -drop and also overlaps with the faradaic response [15*,18,23]. Capabilities, technical details and validation of the simulator are described in Sections S1 and S2 of the [Supporting Information](#), and theoretical aspects are available by consulting the extensive literature [24**25–29,30*,31,32**,33–41].

Using MECSim Input parameters

Figure 1 shows a sample of the input file for a simple electron transfer reaction of the form



with $E^0 = 0.000$ V, $k^0 = 2.0 \times 10^{-2}$ cm s⁻¹ and $\alpha = 0.50$. Where E^0 is the reversible potential, k^0 is the heterogeneous electron transfer rate constant at E^0 and α is the charge transfer coefficient with the Butler–Volmer model of electron transfer assumed to apply. Black denotes the basic parameters at the top of the left column and the advanced ramp settings at the bottom of the column. Parameters in black on the top of the right column allow the user to select the type of electrode geometry and associated parameters. Blue denotes technical parameters that generally should not be altered. Red denotes parameters for setting concentrations of species, both in solution and confined to the surface. Magenta denotes the ac signal to add, while green denotes the background capacitor coefficients. The bold black text at the bottom of the right column sets up the reaction network to be solved — it is described in more detail in the next section. A table providing the units for all parameters is given in Section S3 of the [Supporting Information](#).

Adding chemical reactions

Figure 2 demonstrates the analysis of the two basic types of mechanism when electron transfer is coupled to chemical reactions represented by



or



A new reaction is introduced to the reaction network by adding a new line to the input file. Each line has the

Figure 1

Example Value	! Parameter Input	Example Value	! Parameter Input
298	! Temperature (K)	1	! Geometry type (1=planar, 2=spherical, 3=cylinder, 4=RDE)
50.0	! uncompensated resistance (ohms)	0.070	! Planar: surface area of electrode (cm ²)
0.50	! E_start (V)	1.0e0	! Number of spheres (can be fractional)
-0.50	! E_rev (V)	1.0e-4	! Radius of single sphere (cm)
1	! number of cycles	0.50e0	! Number of cylinders (can be fractional)
0.07451	! scan speed (V/s)	0.001e0	! Radius of single cylinder (cm)
16	! 2 ^N points in time across n cycles (int)	0.10e0	! Length of single cylinder (cm)
1	! correct vs can and freq for DigiPot/FFT (1 = yes, 0 for no)	100.0	! Spatial Resolution (>20)
1	! output type: 0=E,i,t; 1=DigiPot compatible	1.0e-1	! RDE radius (cm)
0	! EC type: 0 = Butler-Volmer, 1 = Marcus theory	1.0e2	! RDE rotation speed (rad/s)
0	! Pre-equilibrium switch: 0=stay with user entered, 1 = apply Pre-eqm operation	1.0e-5	! RDE kinematic viscosity (cm ² /s)
0	! fix number of timesteps (1 = yes; 0 = no)	1	! number of AC sources to add (keep 1 with zero amplitude if want DC only)
1000	! Use a fixed number of timesteps rather than 2 ^N	80.0, 22.0	! AC sin wave: amp (mV), freq(Hz) (REPEAT)
0.10	! beta	2	! number of species (need n lines below)
10.0	! Dstar_min	1.0e-6, 1.0e-5, 0	! REPEAT: initial concentration of i (mol/cm3 or mol/cm2 if SC), Diffusion coeff for i, D_i [cm2/s], surface confined (=1 for yes, else in solution)
0.005e0	! max voltage step	0.0e-6, 8.0e-6, 0	! REPEAT: initial concentration of i (mol/cm3 or mol/cm2 if SC), Diffusion coeff for i, D_i [cm2/s], surface confined (=1 for yes, else in solution)
25.6e0	! time resolution experimentally to correct vs can/f (us)	2	! maximum a^n term for a capacitance given by C = a_0 + a_1 V + a_2 V^2 + ... + a_n V^n
0	! show debug output files as well as MECSimOutput.txt (1=yes; 0=no)	0.0	! Epzc (Volts)
0	! use advanced voltage ramp (0 = E_start=E_end, 1 = use advanced ramp below)	20.0e-6	! a_0 term (F/cm^2)
1	! number of E_rev lines for advanced ramp (if enter 0 then first E_rev value is the final time)	3.0e-6	! a_1 term (F/cm^2)
0.0	! E_start (V)	2.0e-6	! a_2 term (F/cm^2)
0.0	! E_end (V)	0, -1, 1, 0.0, 0.0, 0.0, 0.5, 0.5	! type, reactions (nsp times): kf, kb, E0 (V), ks (cm/s), alpha for BV or lambda* for MH (1eV)
0.50	! E_rev - REPEAT for more complicated ramps		

Typical input parameter file for MECSim for a simple electron transfer reaction of the form $A + e^- \rightleftharpoons B$. Headings for the example values and parameter inputs do not appear in the original text file input, and everything is in a single column rather than two as in this figure. (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

Figure 2

Example Value	! Parameter Input
4	! number of species (need n lines below)
1.0e-6, 1.0e-5, 0	! REPEAT: initial concentration of i (mol/cm3 or mol/cm2 if SC), Diffusion coeff for i, D_i [cm2/s], surface confined (=1 for yes, else in solution)
0.0e-6, 8.0e-6, 0	! REPEAT: initial concentration of i (mol/cm3 or mol/cm2 if SC), Diffusion coeff for i, D_i [cm2/s], surface confined (=1 for yes, else in solution)
0.0e-6, 8.0e-6, 0	! REPEAT: initial concentration of i (mol/cm3 or mol/cm2 if SC), Diffusion coeff for i, D_i [cm2/s], surface confined (=1 for yes, else in solution)
0.0e-6, 8.0e-6, 0	! REPEAT: initial concentration of i (mol/cm3 or mol/cm2 if SC), Diffusion coeff for i, D_i [cm2/s], surface confined (=1 for yes, else in solution)
2	! maximum a^n term for a capacitance given by C = a_0 + a_1 V + a_2 V^2 + ... + a_n V^n
0.0	! Epzc (Volts)
20.0e-6	! a_0 term (F/cm^2)
3.0e-6	! a_1 term (F/cm^2)
2.0e-6	! a_2 term (F/cm^2)
0, -1, 1, 0, 0, 0.0, 0.0, 0.5, 0.5	! type, reactions (nsp times): kf, kb, E0 (V), ks (cm/s), alpha for BV or lambda* for MH (1eV)
1, 0, -1, 1, 0, 1.0e1, 0.4e1, 0.0, 0.0, 0.0	! type, reactions (nsp times): kf, kb, E0 (V), ks (cm/s), alpha for BV or lambda* for MH (1eV)
2, -1, -1, 1, 1, 1.0e-3, 1.0e-6, 0.0, 0.0, 0.0	! type, reactions (nsp times): kf, kb, E0 (V), ks (cm/s), alpha for BV or lambda* for MH (1eV)

Adding chemical reactions to MECSim input files. Changes to the input file given in Figure 1 are highlighted in yellow. The main changes are two extra lines added for the two new chemical species and two extra lines added at the bottom for the two new reactions. See the text for how to format the reaction input lines. (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

same format with the first symbol specifying the type of reaction: “0” for charge transfer (Equation (1)), “1” for first-order chemical (Equation (2)), and “2” for second-order chemical (Equation (3)). Reactions involving surface confined species use the same format but have the species defined as being surface confined in the red section of Figure 1. The next n columns refer to each chem-

ical species and describe their participation in each reaction, where n is set by the total number of species, $n = 4$ in Figure 2. The participation is specified by a “-1” if it appears on the left-hand side of a given reaction, and “1” if it is on the right-hand side. For the case of multiple electron transfers of the form $A + 2e^- \rightleftharpoons B$, it would be written as “-1” in the “A” column and “2” in the “B” col-

umn. In MECSim, charge transfer reactions are always input in reduction form. The oxidation case can be entered by swapping the species, e.g. $A \rightleftharpoons B + e$ is equivalent to $B + e^- \rightleftharpoons A$.

The next two columns are k_f and k_b , which are used if the reaction is a chemical reaction and ignored if a charge transfer reaction. The final three columns specify E^0 , k^0 and α for Butler–Volmer or λ for Marcus–Hush theory. In the second reaction line (highlighted in Figure 2), the first-order chemical reaction with $k_{1f} = 10 \text{ s}^{-1}$ and $k_{1b} = 4 \text{ s}^{-1}$ is added. This will use the k_f and k_b values and ignore the charge transfer reaction parameters as will the third reaction which is a second-order chemical reaction with $k_{2f} = 10^{-3} \text{ mol}^{-1} \text{ cm}^3 \text{ s}^{-1}$ and $k_{2b} = 10^{-6} \text{ mol}^{-1} \text{ cm}^3 \text{ s}^{-1}$. More examples of how to build up complex mechanisms are provided in the website [42].

Tool kits

A tool kit consisting of visualisation and analysis codes written in python is provided with MECSim [42] and is used to process the text file output. Users do not have to use these codes and are free to use any software tools they wish. This flexibility is one of the key advantages of MECSim. Brief details are given below on some of the tools that are useful for standard analysis.

Fast Fourier transform and envelope analysis tools

To demonstrate the use of a range of tools made available alongside MECSim we will go through the basic analysis of simulated data based on electron transfer parameters given in Figure 1 combined with those for the first-order chemical reaction following electron transfer as in Figure 2. The simulated current as a function of time for this EC process is shown in Figure 3a. Commonly, the first step in the analysis is to perform fast FT over the time series data, isolate a component of interest by applying windowing to the power spectrum [43] (Figure 3b), and use inverse fast FT to obtain the waveform shown in Figure 3c. A running maximum algorithm is used to generate the envelope curve as shown for the fundamental harmonic in Figure 3c (bold black trace) and also for 2nd, 5th and 10th in Figure 3d (other harmonics are available but excluded for clarity). Simulated responses for all harmonics are then compared to experimental data that have been processed in exactly the same manner in a parameter optimisation exercise (see below). The visualisations for this process were created using the python codes `HarmonicSplitter.py` and `SmoothPotFiles.py` available along with MECSim [42].

Comparison tools

Simple parameter optimisation comparison tools are also provided, in particular the `CompareSmoothed.py` code which automatically compares the experimental and simulated data. Different weights for each harmonic can be added if a particular harmonic is of special interest. Use

of MECSim with more sophisticated comparison tools is reported elsewhere [3*,14*,15*,17*,18,23,43]. The optimisation approach adopted to estimate unknown model parameters is very flexible and consists of a least squares analysis for fitting simulation to experiment. The procedure is very easy to automate (see section Comparison to experiment) and can extend into finding the optimal fit of single or multiple parameters.

Scripting MECSim

A typical simulation mechanism run for an EC process using MECSim with 2^{16} points takes less than 20 s on a laptop. The two parameter searches presented in the next section use $40 \times 50 = 2000$ parameter sets, which can be easily undertaken without a cluster computer. Having one helps of course and allows for larger searches, multiparameter fitting and implementation of statistical approaches that address the inverse problem and provide error estimates for each parameter. An example of a bash script and a template file used to make the input file for MECSim is shown in Section S4 of Supporting Information. More complex examples of using scripts to run MECSim for different applications, such as those used in the next section, are provided in the website [42].

Comparison to experiment

To illustrate the principles involved in a parameter optimisation exercise, we provide two representative examples—a simple quasi-reversible one-electron transfer process and a stiff scenario where an electron transfer is coupled to a fast and irreversible chemical reaction. The first example is a well-known reduction of ferricyanide to ferrocyanide in a highly conducting aqueous electrolyte solution according to



The unknown parameters in this case that need to be derived might be E^0 and k^0 , with α assumed to be 0.50, R_u known from a separate EIS experiment and C_{dl} known by manual fitting of the experimental data [14*].

The second example is for reduction of *trans*-stilbene in acetonitrile (0.10 M Bu_4NPF_6 , R_u measured by EIS). The process includes a fast one-electron reduction of *trans*-stilbene to anion-radical (Reaction 5) and reduction of the latter to a dianion (Reaction 6), which is rapidly and irreversibly protonated to bibenzyl in a pseudo first-order chemical reaction 7 [17*,20].

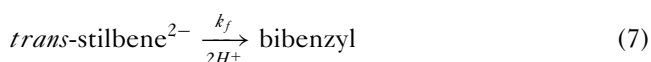
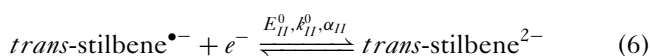
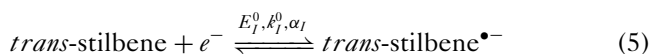
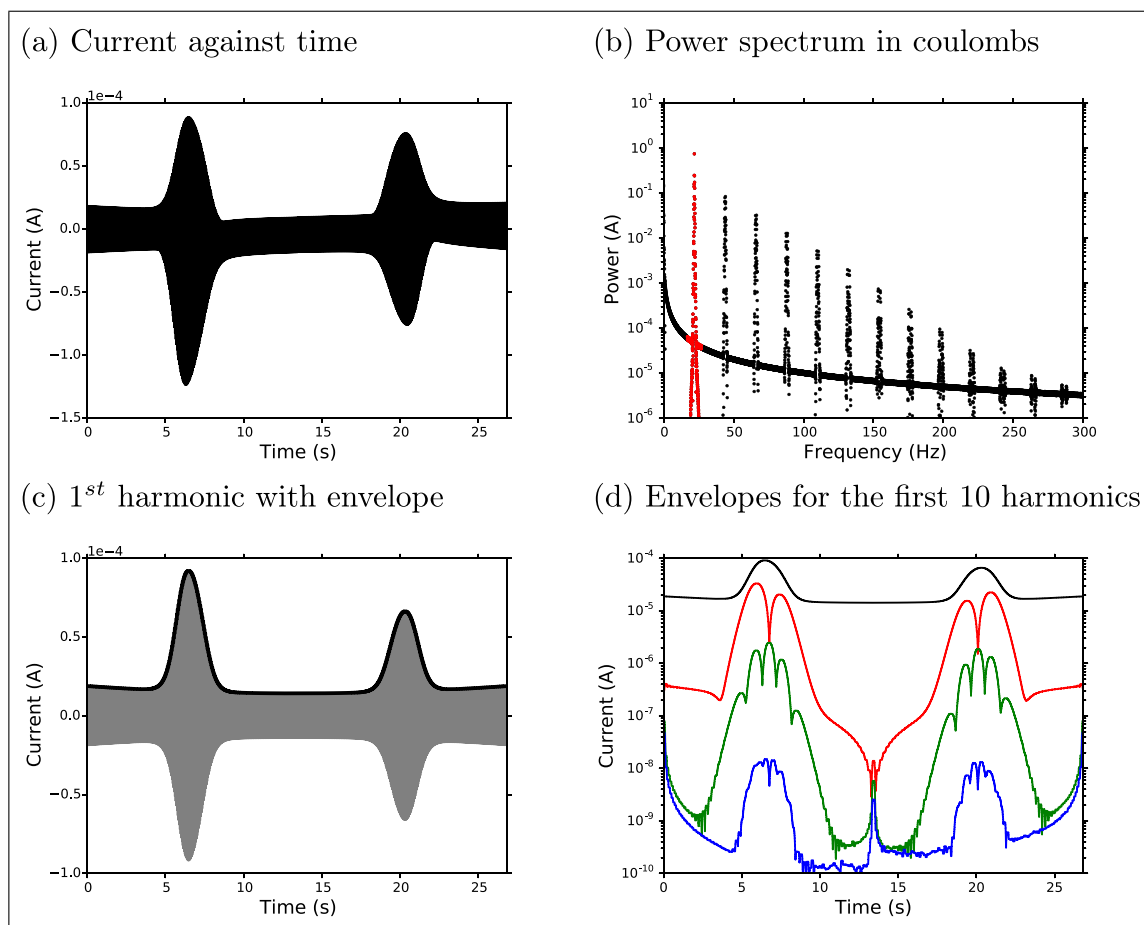


Figure 3



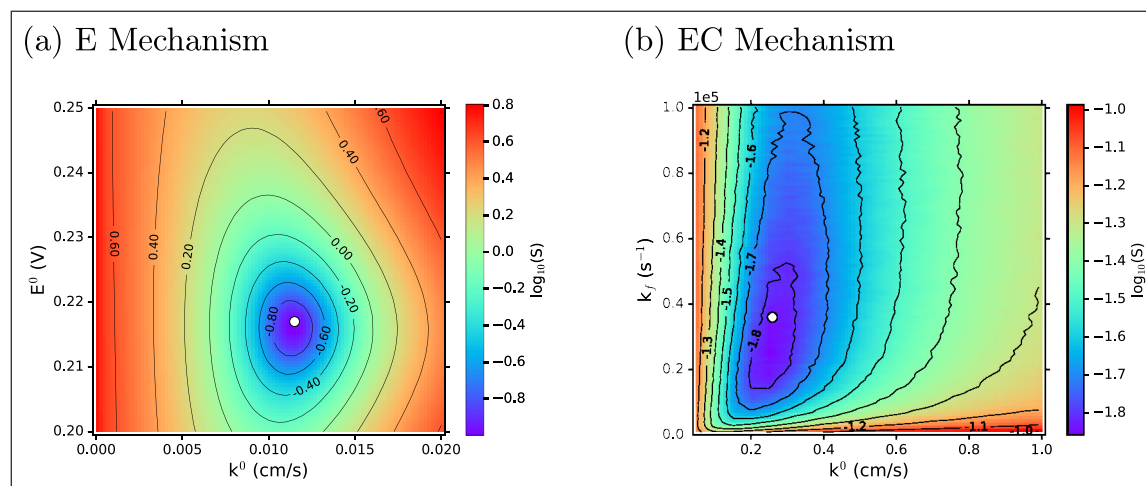
Sequence of steps used to resolve ac voltammograms into aperiodic and ac harmonic components using a Fourier transformation-band filtering-inverse Fourier transformation series of operations. First, the time data (a) are converted to frequency domain via fast Fourier transform (FFT). Next, the harmonic of interest is selected (red in b) and isolated by a windowing exercise [43]. Finally, the band filtered aperiodic component or ac harmonic of interest is converted back into the time domain using an inverse FFT operation (c). When a running maximum algorithm is used to generate the thick black envelope curve, as shown in (c), then resolved envelope forms are generated as displayed for 1st (black), 2nd (red), 5th (green) and 10th (blue) ac harmonics in panel (d) (note the log scale). Example displayed is based on data derived from simulation of an EC mechanism using parameters given in the text. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The parameters for the process in reaction 5 (E_I^0 , k_I^0 , α_I) and E_{II}^0 for reaction 6 can be established from separate experiments [20], while α_{II} is arbitrarily assumed to be equal to $\alpha_I = 0.50$. Thus, two remaining unknowns are k_{II}^0 for reaction 6 and k_f for reaction 7 (k_b is assumed to be infinitely low).

The method for comparisons between experimental and MECSim simulated data adopted here builds upon the analysis tools described above and is as follows:

1. Remove any spurious spikes in the experimental data, e.g. noise or ringing artefacts [43].
2. From the experimental data, obtain the smoothed envelopes against time for the harmonics as shown in Figure 3.
3. Setup a MECSim input file based on fixed and variable parameters.
4. Run the MECSim simulation to generate current against time data for each set of parameters.
5. Apply analysis from step 2 to the simulated data.
6. Use a least squares comparison for each ac harmonic current envelope between the experimental and simulated data.
7. Repeat this process for each combination of the unknown parameters.

Figure 4



Least squares comparison between experimental data and MECSim simulation results for (a) E and (b) EC mechanisms as a function of two unknown parameters: (a) E^0 and k^0 , (b) k^0 and k_f . The values of all other parameters are assumed to be known. The processes considered in this exercise are described in detail in Morris *et al.* [14*] and Simonov *et al.* [17*], respectively. Best fits based on the parameters from the literature are shown as white circles in each case. (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.)

By combining the least squares comparisons between the MECSim model and the experimental data we can build a map of quality of agreement as a function of variable parameters. Figure 4 shows the comparison metric values ($\log_{10}S$) derived for each ac harmonic envelope as

$$S = \sum_{j=1}^{N_{\text{Harm}}} S_j \quad (8)$$

where the scaled least squares value for the envelope of each harmonic is given by

$$S_j = \frac{\sum_{k=1}^N (i_k(t_k) - f_k(t_k))^2}{\sum_{k=1}^N (i_k(t_k))^2} \quad (9)$$

where N is the total number of time steps, i_k is the current at time t_k determined by experiment and f_k is the current calculated by MECSim at the same time.

By scripting MECSim to run across ranges for two parameters the best fit can be found automatically and far faster than the previous “by-eye” (heuristic) method. For the specific cases considered here, the best-fit values for the E mechanism (Figure 4a) were found to be $E^0 = 0.216$ V and $k^0 = 0.0115$ cm s⁻¹, i.e. identical to the values for experiment 10 from Morris *et al.* [14*]. For the EC mechanism (Figure 4b), the best-fit values are $k_f = 2.5 \times 10^4$ s⁻¹ and $k^0 = 0.26$ cm s⁻¹, which fall within the best-fit range from Simonov *et al.* [17*] ($k_f = 3.6 \times 10^4$ s⁻¹ and $k^0 = 0.26$ cm s⁻¹). These comparison examples for two typical mech-

anisms highlight a major benefit of MECSim, namely that it is possible to implement automatic parameter optimisation.

Conclusions

We have introduced, and briefly outlined, the MECSim software package that models ac voltammetry including the large amplitude form that generates a series of harmonics that are accessible by FT processing. The MECSim package can model cases that previous simulation packages cannot, in particular non-linear background capacitance. Where possible we have validated dc data generated by the MECSim simulation package against those from commercially available packages, noting that use of zero amplitude or frequency for the added sine wave as a parameter input allows MECSim to convert to a dc simulation package. We have also tested the new capabilities provided by parameter optimisation underpinned by MECSim calculations on real experimental data. This important attribute is not available with most commercially available software packages. However, of course other packages provide access to 2D simulations, Frumkin and other adsorption isotherms and more accurate simulations of very fast homogeneous reactions coupled to electron transfer which are not available in MECSim.

The authors suggest that limited access to the theory as a reason for not using ac voltammetry is now removed. Ac now like dc voltammetry has advanced simulation model support, which was the missing link. Just as importantly, MECSim can be used automatically via scripts and hence is perfectly suited for cluster/cloud computing. MECSim

is sufficiently fast to run that it can be used for parameter optimisation on a laptop computer when only a modest number of parameters need to be determined.

Acknowledgements

The concept of developing the general ac voltammetric simulation package MECSim was inspired by the works of Stephen Feldberg, who pioneered the introduction of many aspects of digital simulations into the field of electrochemistry including co-developing DigiSim and Manfred Rudolph who has also made many important contributions to simulations of voltammetry as well as being a co-developer of DigiSim and the inventor of DigiElch.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.coelec.2016.12.001](https://doi.org/10.1016/j.coelec.2016.12.001).

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